Postmark: Watercolor Stamp PRD

Oct 21, 2026 · @Janhavi

Overview

Postmark is a mobile website where travellers paint a small watercolor stamp of the place they are in, and keep it as a postmarked souvenir. It opens instantly, works with no signal, and the finished stamp looks like real watercolor on paper.

The moment

You are standing somewhere beautiful, looking at mountains, and you want to make something. A photo feels too easy. You have no sketchbook. Your phone is the only thing in your hand.

Why existing apps miss it

- **Too heavy.** Procreate Pocket, Fresco and Infinite Painter need an install, an account, and a blank full-size canvas. That is a studio, not a five-minute pause.
- **Too hard on a small screen.** A full painting with a finger on a 6-inch screen usually looks bad, and people stop.
- **No sense of place.** The output is just an image. Nothing ties it to where and when you made it.

The bet

Shrink the canvas and the ambition. You are not painting a painting, you are painting a stamp. A small canvas is easy to fill, a limited palette makes anyone look good, and the postmark turns it into a memory of a place.

Goals, non-goals and success

Goals

1. A finished stamp looks like real watercolor on paper to someone seeing it on Instagram.
2. Someone with no painting skill can make a stamp they like in under 5 minutes.
3. It works on a phone in a browser, with no install, no account and no signal.
4. The stamp carries where and when it was made.

Non-goals (v1)

- Large or full-screen paintings
- Layers, selection tools, or anything Procreate does
- Choosing a stamp shape: v1 has one stamp design
- An in-app album of saved stamps: v1 is export only, the album comes later
- Accounts, cloud sync, or a public gallery
- Desktop as a main platform (it should work, but phone comes first)

How we know it works

Signal	Target
People who start painting and finish a stamp	More than half
Finished stamps that get exported	More than 70%
Time from opening the site to first stroke	Under 5 seconds
Blind test: real watercolor scan vs Postmark stamp	Testers can't reliably tell which is which

The blind test is the one that matters most. If the texture fails, nothing else saves it.

User and core flow

Who it's for

A traveller in their 20s or 30s, phone in hand, at a viewpoint, a cafe, or on a train. They like making things but don't carry art supplies. They are not a trained painter. They have five to fifteen minutes and maybe no network.

The flow

1. **Open.** The site loads straight into a blank stamp on paper. No splash screen, no sign-up.
2. **Pick a palette.** The app asks for location and suggests the preset that fits the place. The user can take it, pick another preset, use a saved custom palette, or build a new one. Any palette can be edited and extended.
3. **Paint.** Paint with a finger or stylus. The screen stays awake and the page never scrolls or zooms.
4. **Finish.** Tap Done. The painting dries: edges settle and the texture sets.

5. **Postmark.** The place and date fill in from the location picked up at the start. The user can edit both.
6. **Export.** Choose stamp, sticker or plain PNG, then save to Photos or share.

If the user leaves mid-painting (a call, a lock screen), the painting is still there when they come back.

Painting requirements

The brush has to feel and look like watercolor. Everything else is secondary.

Canvas

- A stamp-shaped canvas (portrait, roughly 4:5) centred on screen, with paper showing around it
- Painting is stored at 2x to 3x screen resolution so exports are sharp
- No scroll, no pinch zoom, no pull-to-refresh, no text selection while painting
- Screen stays awake while the canvas is open (Wake Lock)

Tools

- **Brushes:** small, medium and large round brush, plus a wet brush that softens and blends without adding color
- **Pressure:** uses stylus pressure where available (Apple Pencil, some Android pens); on a finger, speed controls thickness (fast is thinner)
- **Palette:** preset or custom, see Palettes below
- **Water amount:** one slider from dry to soaking wet; wetter means more spread and softer edges
- **Undo and redo:** at least 20 steps
- **Clear:** asks before wiping

Palettes

The user starts with a palette but is never stuck with it.

- **Three ways to start:** pick a preset, use a custom palette saved before, or build a new one from scratch
- **Presets by place type, kept to a minimum:** Mountains, Beach, City, Forest, Desert (6 colors each). Location suggests one; with no signal or an unclear place, the user picks
- **Everything is editable:** change any swatch, remove it, or add more colors, up to 12 per palette

- **Save as custom:** an edited preset or a new palette can be saved with a name, stored on the device
- **Mid-painting edits:** the palette can be changed while painting, without losing the painting
- **No white swatch:** white is the paper

Watercolor realism (must have)

Effect	What it looks like	Why it matters
Edge darkening	A darker rim where a stroke dries	The single biggest tell of real watercolor
Glazing	Overlaps get deeper, never opaque	Paint is transparent, layers multiply
Granulation	Pigment settles into the paper's grain	Makes it feel physical, not digital
Bleeding	Wet color spreads in soft, uneven blooms	Wet-on-wet is what people love about watercolor
Paper texture	Real scanned cold-press grain everywhere	White areas must still look like paper
Uneven wash	Slight blotchy variation inside a flat stroke	Perfectly even fill reads as fake

Drying

Strokes stay wet for a few seconds and keep spreading a little, then settle. Painting into a wet area bleeds. Painting over a dry area glazes. This one behaviour makes it feel alive.

Postmark and location

The postmark uses the real location by default, and the user can edit it after the painting is done.

When to ask

Ask for location at the palette step, with a clear reason: "to suggest colors for where you are." The same location fills the postmark at Done. If the user says no, they pick a palette themselves and type the place at the end.

What fills the postmark, in order

1. **Place name** from the phone's location, looked up online (for example "Manali, Himachal Pradesh")
2. **Coordinates** if there is no signal, for example "32.24° N, 77.19° E". The place name is filled in later once the phone is back online, if the user hasn't edited it.
3. **Empty field** for the user to type, if location is denied

Rules

- Town or area level only, never a street or exact spot. Stamps get shared publicly.
- Date comes from the phone, in a postmark style like "21 OCT 2026"
- Place and date are both editable text fields, with a character limit so they fit the circle
- Optional denomination text on the stamp, like "₹5", also editable
- Place lookup uses a free service such as OpenStreetMap's Nominatim for v1

Export

Three formats in v1, all PNG, all previewed before saving.

Format	What it is	Background	Best for
Stamp	Painting with perforated edges, postmark circle and wavy cancel lines over a corner	Transparent outside the perforations	Instagram stories, journals, the main output
Sticker	Painting with a thick white die-cut border and soft shadow	Transparent	WhatsApp and iMessage stickers
Painting only	Just the painting on its paper	Paper	Printing, wallpapers

Requirements

- Export at high resolution, at least 1500 px on the long side
- On phones, use the share sheet ("Save Image", share to WhatsApp) instead of a download link, because downloads on iPhone Safari land in Files and confuse people
- Fall back to a normal download on desktop
- File name includes place and date, like `postmark-manali-2026-10-21.png`

Later

- Postcard format with a handwritten-style note on the back
- An in-app album where your stamps live

Technical requirements

Keep the stack small enough to vibe code with Claude Code, and test on a real phone from day one.

Area	Choice	Note
Setup	Vite, plain JavaScript or React	Vite lets you open the site on your phone over Wi-Fi while building
Painting engine	WebGL shaders, with Canvas 2D for the UI	Shaders do the edge darkening, granulation and bleeding fast enough on phones
Paper	Your own scanned cold-press paper, plus a few scanned washes	Real scans beat stock textures
Touch	Pointer Events, `touch-action: none`	Covers finger, stylus and mouse in one API, with pressure where supported
Screen awake	Wake Lock API	Re-request it when the page comes back into view
Offline	PWA with a service worker	Works with no signal once opened, can be added to the home screen
Saving work	IndexedDB auto-save after each stroke	Survives calls, lock screen and closing the tab
Location	Geolocation API + Nominatim	Name lookup only when online
Sharing	Web Share API, download fallback	Gets the PNG into Photos on iPhone
Hosting	Vercel, free tier	Needs HTTPS anyway for location and Wake Lock

Performance

- Smooth painting (60 fps) on a mid-range Android phone and an iPhone from the last 4 years
- First load under 1 MB before textures, paper texture loads progressively

- Paint simulation runs at a lower resolution while wet, full resolution when it dries

Browsers

Safari on iPhone, Chrome on Android, then desktop Chrome and Safari. iPhone Safari is the hardest and gets tested first.

Shader approach

The watercolor look runs on WebGL fragment shaders, built in small steps and tuned by eye with on-screen sliders.

Pipeline for one stroke

1. **Wet layer.** The stroke is drawn into a hidden framebuffer holding pigment color and water amount per pixel.
2. **Spread.** Each frame, a shader lets water and pigment flow into neighbouring pixels, using two framebuffers that swap (ping-pong). Wetter areas spread more.
3. **Composite.** A second shader draws the wet layer onto the paper, adding edge darkening (wet pixel next to dry pixel gets darker) and granulation (pigment collects in the paper texture's dips).
4. **Dry.** Water evaporates over a few seconds. When a stroke is dry, it is baked into a permanent dry layer, so new strokes glaze over it with multiply blending.
5. **Export.** The dry layer plus paper is rendered once at full resolution for the PNG.

Inputs

- **Textures:** scanned cold-press paper (grain and height), the wet layer, the dry layer
- **Uniforms:** brush color, brush size, water amount, time, and every tuning value below

Tuning sliders (Tweakpane or lil-gui, dev only)

Every visual number is a slider while building, so the look is tuned by eye on a real phone next to real watercolor scans:

- Edge darkening strength and width
- Spread speed and how much water slows it down
- Granulation strength
- Paper grain contrast
- Drying time
- Wash unevenness (noise amount inside a stroke)

Once the look is right, the slider values become fixed defaults and the panel is hidden from users.

Build order

One step at a time, checked on the phone before moving on, because shader bugs usually show up as a blank screen with no error:

1. Paper texture showing on screen
2. A flat stroke on the paper with multiply blending
3. Edge darkening
4. Granulation
5. Spread and drying
6. Export at full resolution

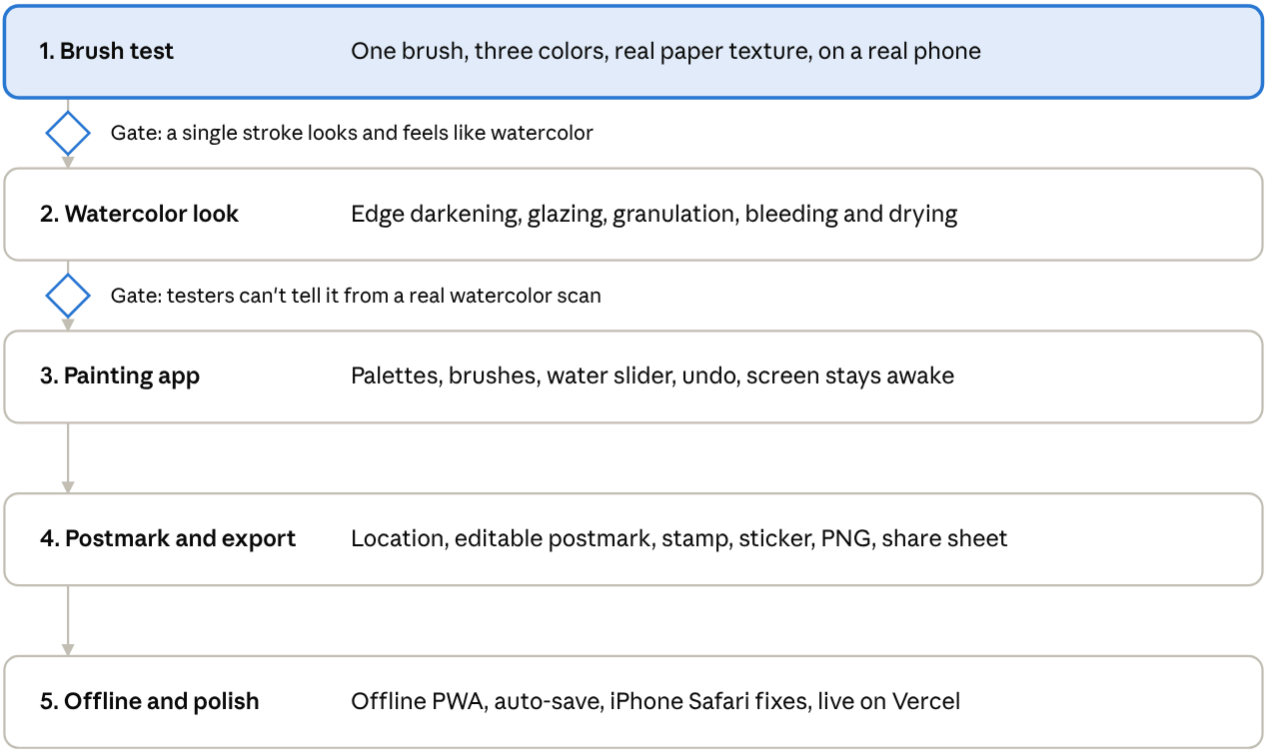
Libraries

A thin WebGL helper such as twgl.js or OGL, not Three.js, which is much bigger than this project needs.

Build plan, risks and decisions

Build in five phases, and don't move past the first two until the brush passes its gates.

Nothing past the brush test gets built until the stroke passes



Phases 1 and 2 carry the risk. Phases 3 to 5 are mostly assembly.

build phases · 5 phases, 2 gates

Phase 1 is a single HTML page tested on your phone. If a stroke doesn't feel right there, no amount of UI will fix it.

Risks

Risk	Fallback
The texture still looks digital	Spend most of the time in Phase 2, compare against your own scans at every step, and keep edge darkening even if bleeding is simplified
Shaders run slow on older phones	Run the wet simulation at lower resolution, turn off bleeding on weak devices
iPhone Safari quirks with Wake Lock, sharing and offline	Test on iPhone first in every phase
Finger painting feels clumsy	Small canvas, speed-based thickness, a wet brush to soften mistakes
Free place lookup has usage limits	Fine for a small project, switch to a paid service if it grows

Decided

- The name is Postmark.
- Palettes: preset, saved custom, or build new. All are editable and extendable.
- One stamp design in v1, no shape choice.
- Export only in v1. The in-app album comes after, if people want it.

Decided in review

- Presets confirmed: Mountains, Beach, City, Forest and Desert.
- 12 colors is the cap per palette.
- The app suggests a preset from the location, which means asking for location at the start instead of at Done.